

AP Calc Lesson 2.8 Homework

Name _____

1. If $f(x) = \frac{e^x}{x^2 + 1}$, then $f'(x) =$

2. Let $f(x) = \frac{x^2 + 4x}{x^2}$.

a. Explain why the quotient rule is not required to find $f'(x)$.

b. Find $f'(x)$

3. If $\frac{d}{dx} \left(\frac{f(x)}{\sin x} \right) = \frac{3x^2 \sin x - x^3 \cos x}{\sin^2 x}$, then $f(x) =$

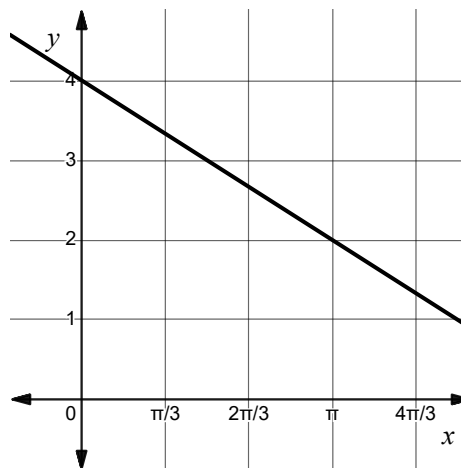
4. The table gives selected values of a differentiable function f . Let g be the function defined by $g(x) = \frac{2 \ln x}{f(x)}$.

x	$f(x)$	$f'(x)$
e	2	6
4	π	-1

a. Find $g'(x)$.

b. What is the value of $g'(e)$?

5. The graph of a function $y = f(x)$ is shown in the figure. If $h(x) = \frac{f(x)}{2 \cos x}$, find $\lim_{x \rightarrow \pi} \frac{h(x) - h(\pi)}{x - \pi}$.

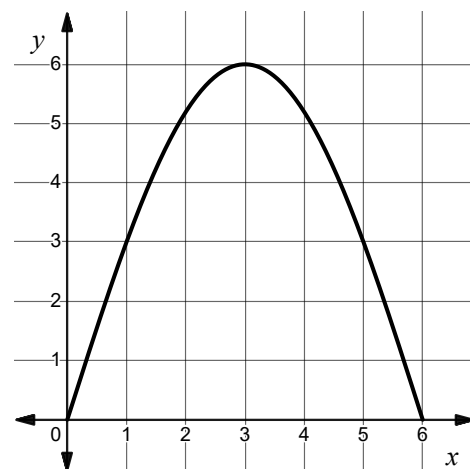


6. The graph of $y = f(x)$ is shown in the figure. Let h be the function given by $h(x) = \frac{f(x)}{13 - 2x}$. For each of the following, indicate whether the derivative is zero, positive, negative, or cannot be determined.

a. $h'(1)$

b. $h'(3)$

c. $h'(5)$



7. Consider the curve given by the equation $y = \frac{x^2}{e^x}$.

a. Find the coordinates of all points on the curve at which the line tangent to the curve is horizontal.

b. Explain why there are no points on the curve at which the line tangent to the curve is vertical.

8. Let function f be given by $f(x) = \frac{x}{x-6}$. Find all values of c such that the line $3x + 8y = c$ is tangent to the graph of function f .

9. The table gives select values of the differentiable functions f and g in terms of a nonzero constant b . These values, along with the first derivatives of f and g , are given for $x = -4$. Let $h(x) = \frac{g(x)}{f(x)}$. Find $h'(-4)$.

x	$f(x)$	$f'(x)$	$g(x)$	$g'(x)$
-4	b	3	$-2b$	5

- A) $-\frac{11}{b}$
- B) $\frac{11}{b}$
- C) $-\frac{1}{b}$
- D) $\frac{5}{3}$
10. Which of the following is an equation for the line tangent to the graph of $y = \frac{x-4}{x^2+2}$ at $x = 1$?
- A) $y = -x$
- B) $y = \frac{1}{2}x - \frac{3}{2}$
- C) $y = x - 2$
- D) $y = 2x - 3$